Naïve Bayes Classifier

WHAT IS BAYESIAN CLASSIFICATION?

- Bayesian classifiers are statistical classifiers
- For each new sample they provide a probability that the sample belongs to a class (for all classes)
- A **Naive Bayes Classifier** is a supervised **machine-learning algorithm** that uses the **Bayes'** theorem, which assumes that features are statistically independent.
- **Bayesian machine learning** allows us to encode our prior beliefs about what those models should look like, independent of what the data tells us.

Type of Naive Bayesian Models

There are three types of Naive Bayes Model, which are given below:

- **Gaussian**: The Gaussian model assumes that features follow a normal distribution. This means if predictors take continuous values instead of discrete, then the model assumes that these values are sampled from the Gaussian distribution.
- **Multinomial**: The Multinomial Naïve Bayes classifier is used when the data is multinomial distributed. It is primarily used for document classification problems, it means a particular document belongs to which category such as Sports, Politics, education, etc. The classifier uses the frequency of words for the predictors.
- **Bernoulli**: The Bernoulli classifier works similar to the Multinomial classifier, but the predictor variables are the independent Booleans variables. Such as if a particular word is present or not in a document. This model is also famous for document classification tasks.

Naive Bayesian Classifier - Application

- It is used for Credit Scoring.
- It is used in medical data classification.
- It can be used in real-time predictions because Naïve Bayes Classifier is an eager learner.
- It is used in Text classification such as Spam filtering and Sentiment analysis.

Naive Bayesian Classifier - Advantages

- Easy to implement
- Good results obtained in most of the cases
- Naïve Bayes is one of the fast and easy ML algorithms to predict a class of datasets.
- It can be used for Binary as well as Multi-class Classifications.
- It performs well in Multi-class predictions as compared to the other Algorithms.
- It is the most popular choice for text classification problems.

Naive Bayesian Classifier - disadvantages

- Assumption: class conditional independence, therefore loss of accuracy
- Naive Bayes assumes that all features are independent or unrelated, so it cannot learn the relationship between features.
- Practically, dependencies exist among variables
  - ✓E.g., hospitals: patients: Profile: age, family history, etc.

    Symptoms: fever, cough etc., Disease: lung cancer, diabetes, etc.
  - ✓Dependencies among these cannot be modeled by Naïve Bayesian Classifier

Bayes' Theorem: Basics

- Let **X** be a data sample (*"evidence"*): class label is unknown
- Let H be a *hypothesis* that X belongs to class C
- Classification is to determine **P(H|X)**, the probability that the hypothesis holds given the observed data sample **X**

Bayes' Theorem: Basics

- In other words, we are looking for the probability that tuple ***X*** belongs to class *C, given that we know the attribute description of* ***X.***
- P(H|X) is the posterior probability, or a posteriori probability, of H conditioned on X.
  - ✓ For example, suppose our world of data tuples is confined to customers described by the attributes *age and income*
  - ✓ X is a 35-year-old customer with an income of $40,000.
  - ✓ P(H|X) reflects the probability that customer X will buy a computer given that we know the customer's age and income

Bayes' Theorem: Basics

- P(H) (*prior probability*), the initial probability
  - ✓ E.g., **X** will buy computer, regardless of age, income, …
- P(**X**): probability that sample data is observed
- P(**X**|H) (*likelihood*), the probability of observing the sample **X**, given that the hypothesis holds
  - ✓ E.g., Given that **X** will buy computer, the prob. that X is 31..40, medium income

Bayes' Theorem

- Given training data **X**, *posteriori probability of a hypothesis* H, P(H|**X**), follows the Bayes theorem,

$$P(H|\mathbf{X}) = \frac{P(\mathbf{X}|H)P(H)}{P(\mathbf{X})}$$

Naïve Bayesian Classifiers

- Let, D be a training set of tuples and their associated class labels,
- Each tuple is represented by an n-D attribute vector

$$\mathbf{X} = (x_1, x_2, \ldots, x_n)$$

- Suppose there are *m* classes $C_1, C_2, \ldots, C_m$.
- Classification is to derive the maximum posteriori, i.e., the maximal $P(C_i|X)$

Naïve Bayesian Classifiers

- This can be derived from Bayes' theorem

$$P(C_i|\mathbf{X}) = \frac{P(\mathbf{X}|C_i)P(C_i)}{P(\mathbf{X})}$$

- Since P(X) is constant for all classes, only

$$P(C_i|\mathbf{X}) = P(\mathbf{X}|C_i)P(C_i)$$ needs to be maximized

NBC: Training Dataset

Class:

C1:buys_computer = 'yes'

C2:buys_computer = 'no'

Data sample

X = (age <=30,

Income = medium,

Student = yes

Credit_rating = Fair

buys_computer =?)

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

NBC: Training Dataset

$$P(C_1|\mathbf{X}) = \frac{P(\mathbf{X}|C_1)P(C_1)}{P(\mathbf{X})}$$

$$P(C_2|\mathbf{X}) = \frac{P(\mathbf{X}|C_2)P(C_2)}{P(\mathbf{X})}$$

Class:

C1:buys_computer = 'yes'=9

C2:buys_computer = 'no'=5

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

NBC: Training Dataset

$P(C_i)$:
P(buys_computer = "yes")
= 9/14
= 0.643

P(buys_computer = "no")
= 5/14
= 0.357

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

NBC: Training Dataset

Data sample

X = (age <=30,
Income = medium,
Student = yes
Credit_rating = Fair)

- ✓ buys_computer = 'yes'=9
- ✓ (age<=30) =2
- ❑ buys_computer = 'no'=5
- ❑ (age<=30) =3

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

NBC: Training Dataset

Data sample

X = (age <=30, Income = medium, Student = yes, Credit_rating = Fair)

- $P(C_i)$: P(buys_computer = "yes") = 9/14 = 0.643
  P(buys_computer = "no") = 5/14= 0.357

- Compute $P(X|C_i)$ for each class
  P(age = "<=30" | **buys_computer = "yes")** = 2/9 = 0.222
  P(age = "<= 30" | **buys_computer = "no")** = 3/5 = 0.6

NBC: Training Dataset

Data sample

X = (age <=30, Income = medium, Student = yes ,Credit_rating = Fair)

- $P(C_i)$: P(buys_computer = "yes") = 9/14 = 0.643

  P(buys_computer = "no") = 5/14= 0.357

- Compute $P(X|C_i)$ for each class

  P(age = "<=30" | buys_computer = "yes") = 2/9 = 0.222

  P(age = "<= 30" | buys_computer = "no") = 3/5 = 0.6

  P(income = "medium" | buys_computer = "yes") = 4/9 = 0.444

  P(income = "medium" | buys_computer = "no") = 2/5 = 0.4

NBC: Training Dataset

Data sample

X = (age <=30, Income = medium, Student = yes ,Credit_rating = Fair)

- $P(C_i)$: P(buys_computer = "yes") = 9/14 = 0.643

  P(buys_computer = "no") = 5/14= 0.357

- Compute $P(X|C_i)$ for each class

  P(age = "<=30" | buys_computer = "yes") = 2/9 = 0.222

  P(age = "<= 30" | buys_computer = "no") = 3/5 = 0.6

  P(income = "medium" | buys_computer = "yes") = 4/9 = 0.444

  P(income = "medium" | buys_computer = "no") = 2/5 = 0.4

  P(student = "yes" | buys_computer = "yes) **= 6/9** = 0.667

  P(student = "yes" | buys_computer = "no") = **1/5** = 0.2

NBC: Training Dataset

Data sample

X = (age <=30,
Income = medium,
Student = yes
,Credit_rating = Fair)

- $P(C_i)$: P(buys_computer = "yes") = 9/14 = 0.643
  P(buys_computer = "no") = 5/14= 0.357
- Compute $P(X|C_i)$ for each class

P(age = "<=30" | buys_computer = "yes") = 2/9 = 0.222

P(age = "<= 30" | buys_computer = "no") = 3/5 = 0.6

P(income = "medium" | buys_computer = "yes") = 4/9 = 0.444

P(income = "medium" | buys_computer = "no") = 2/5 = 0.4

P(student = "yes" | buys_computer = "yes) = 6/9 = 0.667

P(student = "yes" | buys_computer = "no") = 1/5 = 0.2

P(credit_rating = "fair" | buys_computer = "yes") = 6/9 = 0.667

P(credit_rating = "fair" | buys_computer = "no") = 2/5 = 0.4

NBC: An Example

$$P(C_1|\mathbf{X}) = \frac{P(\mathbf{X}|C_1)P(C_1)}{P(\mathbf{X})}$$

- X = (age <= 30 , income = medium, student = yes, credit_rating = fair)

$P(X|C_i)$: $P(X|buys_computer = \text{“yes”}) = 0.222 \times 0.444 \times 0.667 \times 0.667$

$= 0.044$

$P(X|buys_computer = \text{“no”}) = 0.6 \times 0.4 \times 0.2 \times 0.4$

$= 0.019$

NBC: An Example

$$P(C_1|\mathbf{X}) = \frac{P(\mathbf{X}|C_1)P(C_1)}{P(\mathbf{X})}$$

- X = (age <= 30 , income = medium, student = yes, credit_rating = fair)

$P(X|C_i)$: P(X|buys_computer = "yes") = 0.222 x 0.444 x 0.667 x 0.667

= 0.044

P(X|buys_computer = "no") = 0.6 x 0.4 x 0.2 x 0.4

= 0.019

$P(X|C_i)*P(C_i)$:

P(X|buys_computer = "yes") * P(buys_computer = "yes") = 0.044 x 0.643

=0.028

P(X|buys_computer = "no") * P(buys_computer = "no") = 0.019* 0.357

= 0.0067

Therefore, X belongs to class ("buys_computer = yes")

Naive Bayesian Classifier Example

play tennis?

To classify a new sample X:

outlook = sunny
temperature = cool
humidity = high
windy = false
Class=?

Outlook	Temperature	Humidity	Windy	Play Class
sunny	hot	high	false	N
sunny	hot	high	true	N
overcast	hot	high	false	P
rain	mild	high	false	P
rain	cool	normal	false	P
rain	cool	normal	true	N
overcast	cool	normal	true	P
sunny	mild	high	false	N
sunny	cool	normal	false	P
rain	mild	normal	false	P
sunny	mild	normal	true	P
overcast	mild	high	true	P
overcast	hot	normal	false	P
rain	mild	high	true	N

Naive Bayesian Classifier Example

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Outlook	Temperature	Humidity	Windy	Class
overcast	hot	high	false	P
rain	mild	high	false	P
rain	cool	normal	false	P
overcast	cool	normal	true	P
sunny	cool	normal	false	P
rain	mild	normal	false	P
sunny	mild	normal	true	P
overcast	mild	high	true	P
overcast	hot	normal	false	P

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Outlook	Temperature	Humidity	Windy	Class
sunny	hot	high	false	N
sunny	hot	high	true	N
rain	cool	normal	true	N
sunny	mild	high	false	N
rain	mild	high	true	N

Naive Bayesian Classifier Example

- Given the training set, we compute the probabilities:

Outlook	P	N
sunny	2/9	3/5
overcast	4/9	0
rain	3/9	2/5
Tempreature		
hot	2/9	2/5
mild	4/9	2/5
cool	3/9	1/5

Humidity	P	N
high	3/9	4/5
normal	6/9	1/5
Windy		
true	3/9	3/5
false	6/9	2/5

To classify a new sample X:

outlook = sunny
temperature = cool
humidity = high
windy = false
Class=?

- We also have the probabilities
  - P = 9/14
  - N = 5/14

Naive Bayesian Classifier - Assignment

Q1: To classify a new sample X:

- outlook = sunny
- temperature = cool
- humidity = high
- windy = false

- Therefore X takes class label N

Naive Bayesian Classifier - Assignment

Q2: Second example X = <rain, hot, high, false>

- outlook = rain
- temperature = hot
- humidity = high
- windy = false

- Sample X is classified in class N (don't play)

The End